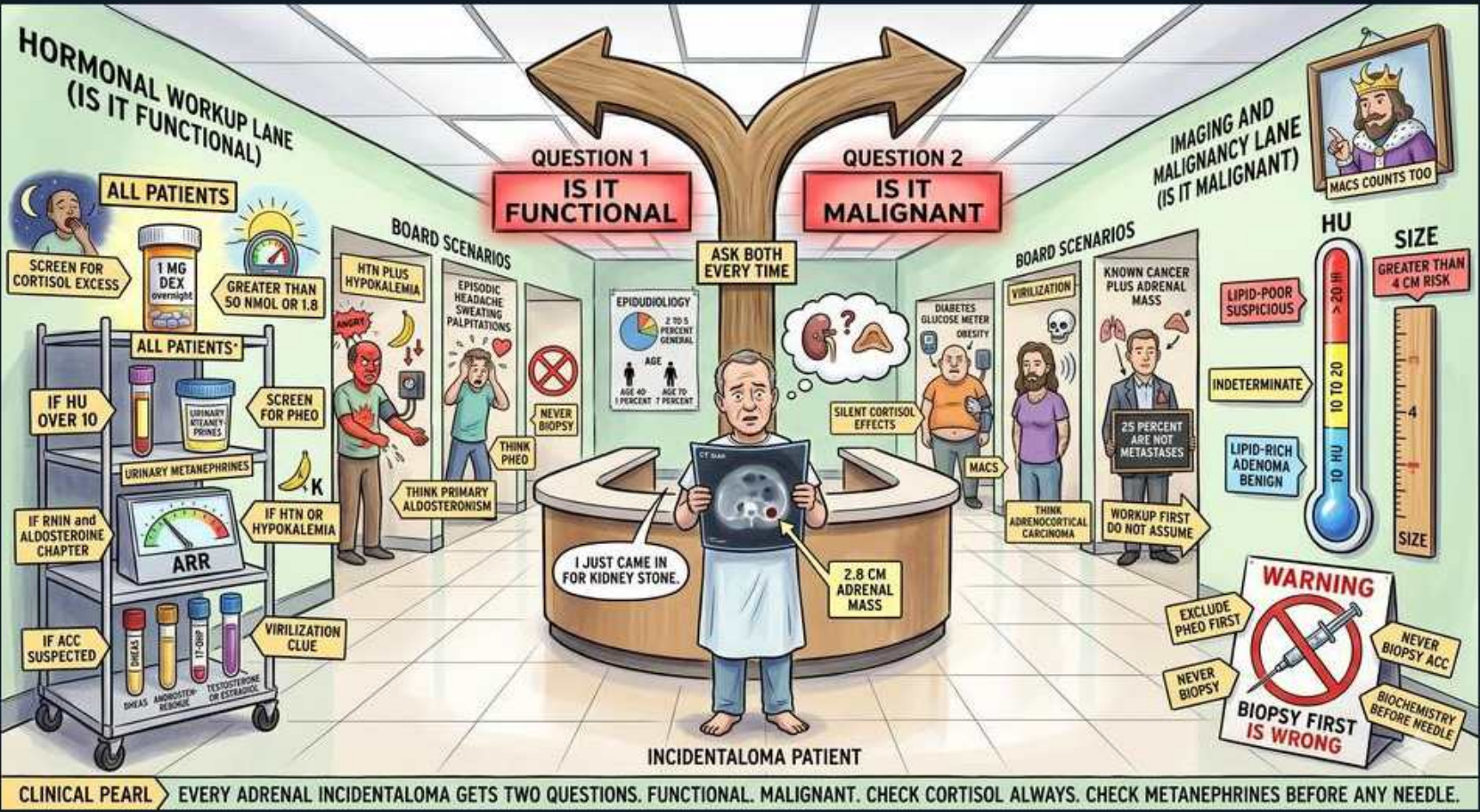


Adrenal — Incidentaloma Triage Bay (Functional? Malignant?)





1. The Two-Question Fork (ceiling sign)

WHAT YOU SEE

A wooden forked arrow on the center ceiling splits left to 'Question 1: Is it Functional?' and right to 'Question 2: Is it Malignant?', with a plate reading 'Ask Both Every Time.'

WHAT IT ENCODES

Every adrenal incidentaloma triggers two simultaneous questions: (1) Is it hormonally functional? and (2) Is it malignant? You never answer one and skip the other. A mass that looks benign on CT still gets a hormonal screen, and an asymptomatic patient still gets a malignancy assessment. Both lanes open at the same time.

BOARD TRAP

Do NOT sequence them — do not wait for imaging before ordering labs, and do not skip labs because the mass 'looks like an adenoma.' Functional and malignant workups run in parallel, every time.

2. Question 1 — Is it Functional? (left banner)

WHAT YOU SEE

The left arm of the fork carries a red banner: 'Question 1 — Is it Functional?'

WHAT IT ENCODES

The functional question is answered with a universal hormonal screen for ALL patients: a 1 mg overnight dexamethasone suppression test (cortisol autonomy) plus plasma or urinary metanephrines (pheochromocytoma). Aldosterone:renin ratio and androgens are added only conditionally.

BOARD TRAP

'No Cushingoid features, no headaches, normotensive — so no hormone workup.' Wrong: the functional screen is mandatory regardless of how the patient looks.

3. Question 2 — Is it Malignant? (right banner)

WHAT YOU SEE

The right arm of the fork carries a red banner: 'Question 2 — Is it Malignant?'

WHAT IT ENCODES

The malignant question is answered by imaging phenotype and size, not by biopsy. Unenhanced CT attenuation and a 4 cm size threshold drive risk stratification for adrenocortical carcinoma (ACC) or metastasis. Functional lesions, masses >4 cm, or suspicious imaging go to adrenalectomy; the rest get surveillance.

BOARD TRAP

Reaching for a needle to 'find out if it's cancer.' Malignancy is assessed by imaging criteria and resected surgically — never biopsied to make the call.

4. Incidentaloma patient at triage desk

WHAT YOU SEE

A man at the triage desk holds a CT film with a yellow-arrowed 2.8 cm adrenal mass; his speech bubble says 'I just came in for a kidney stone.'

WHAT IT ENCODES

Defines an incidentaloma: an adrenal mass (usually >1 cm) found by imaging ordered for an unrelated reason, in a patient with no adrenal symptoms. The mass was never being looked for, yet it still mandates the full two-question triage.

BOARD TRAP

Mistaking 'no symptoms' for 'no workup needed.' Absence of Cushing, virilization, or hypertension does not exempt the patient from the standard triage.

5. Epidemiology pie chart (age matters)

WHAT YOU SEE

A small pie chart under the fork reads '2–5% general population, 7% by age 70' with 'Age 40: 1%' and 'Age 70: 7%.'

WHAT IT ENCODES

Incidentalomas are common (2–5% of the population) and rise with age (~1% at 40 to ~7% at 70). Most (40–70%) are benign non-functional cortical adenomas, but MACS (mild autonomous cortisol secretion) accounts for 20–50% — easily missed if you only screen overtly symptomatic patients.

BOARD TRAP

'It's just a benign adenoma in an elderly patient, no workup.' Age raises prevalence but never changes the workup mandate.

6. Universal hormonal workup cart — '1 mg DST + metanephrines'

WHAT YOU SEE

A cart labeled 'All Patients' on the upper-left holds a dexamethasone pill bottle ('Screen for cortisol excess') and a urinary metanephrines tube ('Screen for pheo').

WHAT IT ENCODES

Two tests for EVERY incidentaloma patient: (1) 1 mg overnight dexamethasone suppression test — detects autonomous cortisol secretion / MACS; post-DST cortisol >1.8 µg/dL (50 nmol/L) is abnormal. (2) Plasma or 24-h urinary metanephrines — excludes pheochromocytoma. Pheo must be excluded before any biopsy or surgery.

BOARD TRAP

'No hypertension or headache, so skip the pheo screen.' Wrong — metanephrines are universal and must be done BEFORE any needle or operation.

7. Conditional tests cart — ARR gauge & androgens

WHAT YOU SEE

Lower on the same cart, branching lanes show an 'ARR' gauge ('if renin/aldosterone' and 'if HTN or hypokalemia'), a 'virilization clue' androgen rack, and 'if ACC suspected.'

WHAT IT ENCODES

Conditional, not universal, tests: aldosterone:renin ratio (ARR) only if hypertension or hypokalemia (screens for primary aldosteronism/Conn). Androgens (DHEAS, androstenedione, testosterone/estradiol) only if virilization. Add 17-OHP to the androgen panel when ACC is suspected.

BOARD TRAP

Ordering ARR on every incidentaloma patient. ARR is triggered ONLY by HTN or hypokalemia — blind ordering inflates false positives.

8. Functional red-flag doors (aldosteronism & pheo)

WHAT YOU SEE

Two left-wall doors: an angry red hypertensive patient holding a banana ('Think Primary Aldosteronism'), and a sweating patient with palpitations ('Think Pheo — Never Biopsy').

WHAT IT ENCODES

The two most board-tested functional syndromes: HTN + hypokalemia (the banana) = primary aldosteronism (Conn) → confirm with ARR. Episodic headache + sweating + palpitations = pheochromocytoma → confirm with metanephrines. The 'Never Biopsy' tag on the pheo door is the critical warning.

BOARD TRAP

Biopsying a suspected pheo before checking metanephrines can precipitate a fatal hypertensive/catecholamine crisis. Biochemistry before needle — always.



9. MACS — Cortisol King portrait (upper-right)

WHAT YOU SEE

A framed portrait of a crowned king (Cortisol) points outward, labeled 'MACS Counts Too.'

WHAT IT ENCODES

MACS (mild autonomous cortisol secretion) is subclinical Cushing: no cushingoid features, but cortisol autonomy proven on DST. Post-DST cortisol 1.8–5.0 µg/dL = possible/MACS; ≥5.0 µg/dL = confirmed autonomous secretion. It still drives real morbidity — type 2 diabetes, hypertension, osteoporosis, cardiovascular events — and may warrant adrenalectomy if comorbidities are present.

BOARD TRAP

'Post-DST cortisol is 2.1 µg/dL, so the patient is normal.' Wrong — ≥1.8 µg/dL already signals autonomous secretion. MACS is defined by the DST number, not by how the patient looks.

10. Imaging & Malignancy lane label (right wall)

WHAT YOU SEE

A wall banner on the right reads 'Imaging and Malignancy Lane — Is It Malignant?'

WHAT IT ENCODES

The malignancy lane is driven by CT phenotype and size, plus context. Key features favoring malignancy: large size (>4 cm), lipid-poor high attenuation (>20 HU), poor contrast washout, irregular margins, and rapid growth. The two malignant entities are ACC (primary) and metastasis.

BOARD TRAP

Assuming any large or heterogeneous adrenal mass is automatically a met. Phenotype must be characterized — and 25% of adrenal masses in cancer patients are NOT metastases.

11. Virilization & known-cancer doors (right lane)

WHAT YOU SEE

Two right-lane doors: a bearded, masculinized patient holding a 'Known Cancer Plus Adrenal Mass' sign, and a patient labeled '25% Are Not Metastases — Workup First.'

WHAT IT ENCODES

Known cancer + adrenal mass does NOT equal metastasis — about 25% (1 in 4) are primary adrenal tumors (adenoma or ACC), so full workup still applies. Virilization in a woman with an adrenal mass (hirsutism, clitoromegaly, deep voice) = adrenocortical carcinoma until proven otherwise.

BOARD TRAP

'Patient has lung cancer and an adrenal mass, so it must be a met.' Wrong — work it up, exclude a functional/pheo lesion, and characterize it on imaging first.

12. HU thermometer (Hounsfield units)

WHAT YOU SEE

A thermometer labeled 'HU' shows zones: '≤10 = lipid-rich adenoma (benign)', '10–20 = indeterminate', '>20 = lipid-poor / suspicious.'

WHAT IT ENCODES

On UNENHANCED CT: ≤10 HU = lipid-rich adenoma, almost certainly benign, no further imaging. 10–20 HU = indeterminate → washout CT or chemical-shift MRI. >20 HU = lipid-poor = suspicious for malignancy. On washout CT, absolute washout >60% (or relative >40%) indicates a benign adenoma.

BOARD TRAP

Using HU cutoffs in isolation. A mass can be lipid-rich yet still worrisome by size — HU and size must always be read together.

13. Size ruler — 4 cm cutoff

WHAT YOU SEE

A ruler beside the thermometer marks the 4 cm line in red: 'Size >4 cm = Risk.'

WHAT IT ENCODES

Size >4 cm independently raises concern for adrenocortical carcinoma regardless of attenuation. Functional masses, masses >4 cm, or imaging-suspicious masses are referred for adrenalectomy; smaller, benign-appearing, non-functional masses go to surveillance.

BOARD TRAP

Reassuring on a 5 cm mass because it measured 8 HU. Size >4 cm warrants concern even when attenuation looks benign — both parameters count.

14. 'Biopsy First Is Wrong' warning sign

WHAT YOU SEE

A large red warning sign with a crossed-out needle reads 'Biopsy First Is Wrong,' flanked by 'Exclude Pheo First,' 'Never Biopsy ACC,' and 'Biochemistry Before Needle.'

WHAT IT ENCODES

Adrenal biopsy has two absolute contraindications: (1) pheo not yet excluded — biopsy can trigger a catecholamine crisis; (2) suspected ACC — biopsy risks tumor seeding and forfeits surgical cure. The ONLY valid indication is confirming an adrenal metastasis from a known extra-adrenal primary, and only after pheo is biochemically excluded.

BOARD TRAP

Biopsying an adrenal mass to 'determine if it's malignant.' Never biopsy to assess malignancy — that is the job of imaging criteria and surgery; and never biopsy until metanephrines are negative.

15. Clinical pearl banner (bottom)

WHAT YOU SEE

The bottom banner reads: 'Every adrenal incidentaloma gets two questions. Functional. Malignant. Check cortisol always. Check metanephrines before any needle.'

WHAT IT ENCODES

The single sentence governing the scene: the two-question triage (functional? + malignant?) is universal, the cortisol/DST screen is universal, and pheo exclusion via metanephrines must precede any invasive procedure. Never biopsy to assess malignancy, and never assume a cancer patient's adrenal mass is a metastasis.

BOARD TRAP

Forgetting that the pearl is absolute: cortisol always, metanephrines before any needle — these are not optional steps that depend on symptoms.